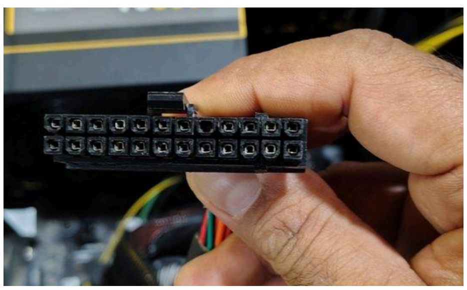
Task 4 - Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Mount the SMPS and route the cables

Cable / connector	

Before connecting anything, the SMPS has to be mounted in the case with its fan facing the vented panel, and all cable bundles were routed through the back of the case to keep the front tidy and improve airflow.

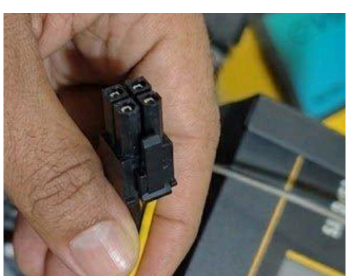
Step 2: Connect the main motherboard power



Cable / connector	ATX 24-pin connector

The 24-pin ATX connector was plugged into the matching socket on the motherboard edge. It was pushed in firmly until the side clip clicked, since this single connector supplies the +3.3V, +5V, and +12V rails the motherboard and chipset need to power on.

Step 3: Connect the CPU 12V power



Cable / connector	CPU ATX 4-pin 12V connector (EPS)

The 4-pin CPU connector was plugged into its dedicated socket near the CPU socket on the motherboard, next to the cooler fan. This supplies focused +12V power just for the processor, separate from the main 24-pin rail, and its small locking clip was checked to make sure it had clicked in fully.

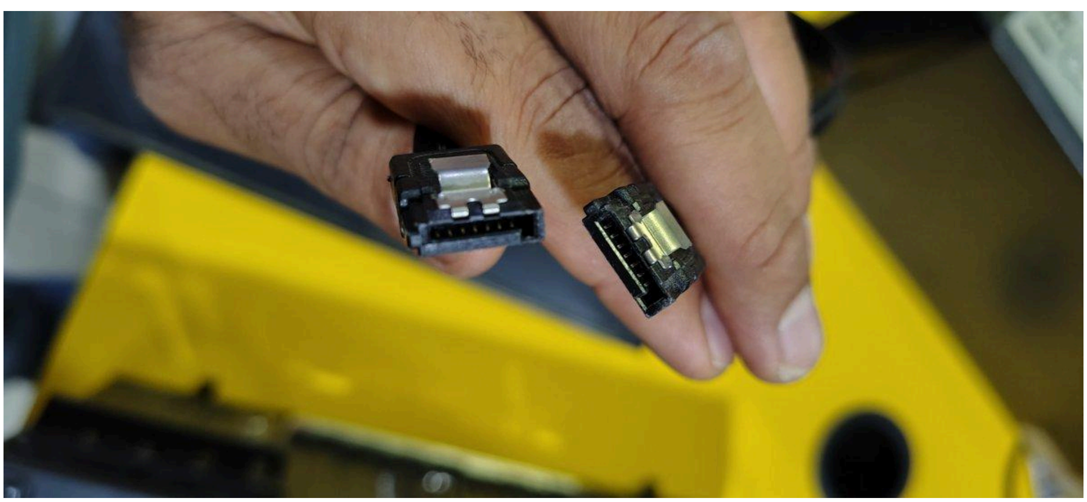
Step 4: Connect storage drives — power

Cable / connector	SATA power connector

The flat, L-shaped SATA power connector (orange/red/yellow/black wires) was connected to the SSD/HDD's power input. The keyed shape only allows it to go in one way, so it was gently pushed in until it clicked into place — no force was needed.

A second SATA power connector from the same cable run was used for an additional drive, as shown in the close-up of the connector pins below.

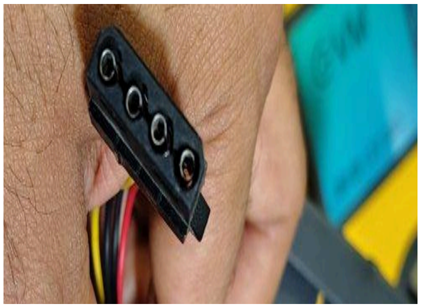
Step 5: Connect storage drives — data



Cable / connector	SATA data connector (7-pin)

Alongside the power cable, the thinner SATA data connector was plugged into the drive and routed to the motherboard's SATA port. This is a separate cable from SATA power — it carries the actual data signal, not electricity — and its pins are more fragile, so it was seated gently along the keyed notch rather than forced.

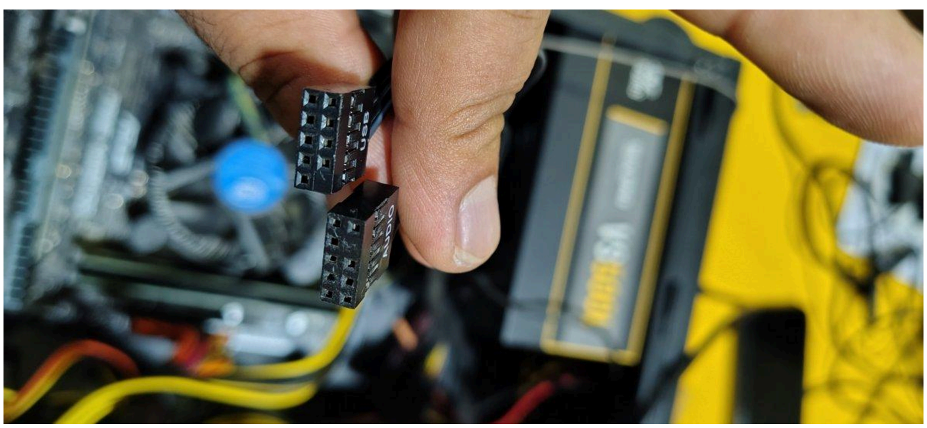
Step 6: Connect any Molex-powered accessories



Cable / connector	Molex 4-pin connector

A 4-pin Molex connector from the PSU was used to power an additional case fan. The bevelled corners were lined up first so the connector could only go in the correct way round, preventing a reversed-polarity connection.

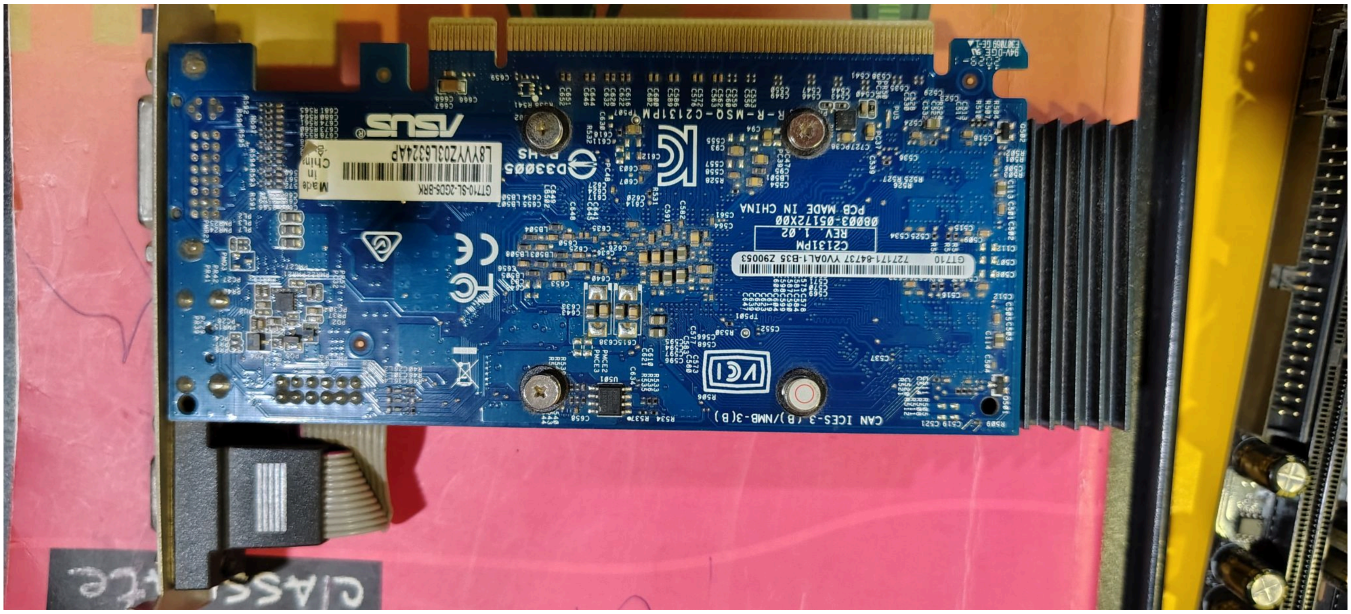
Step 7: Wire the front-panel USB and audio headers



Cable / connector	USB header and AUDIO header (motherboard front-panel pins)

These two connectors are not PSU power cables but front-panel headers — they were connected from the case's front USB ports and headphone/mic jack to the labelled USB and AUDIO pin blocks on the motherboard, completing the case wiring alongside the SMPS connections.

Step 8: Connect the graphics card power



Cable / connector	PCIe x16 slot (no separate power connector needed for this card)

The graphics card used in this build is an ASUS GT710, a low-power card that draws everything it needs directly from the PCIe x16 slot on the motherboard — there is no 6-pin or 8-pin PCIe power connector on the card's edge, so no extra SMPS cable was needed for it.

Step 9: Final check and power-on

Cable / connector	All connectors above

With the ATX 24-pin, CPU 4-pin, SATA power, SATA data, Molex, front-panel headers, and the graphics card all seated and clipped in, every connection was visually rechecked for full insertion before closing the case, plugging the SMPS into the wall socket, and powering on the system to confirm everything booted correctly.